ATM Team Round

Paly Math Club

2025-26

DO NOT FLIP THE PAGE UNTIL INSTRUCTED

- No materials are allowed, except for blank scratch paper, pens, pencils, and erasers.
- In particular, no calculators, no electronics, and no cheating.
- You should discuss with your team but may not discuss with others. (Teams can have up to 4 members).
- You may ask board members for clarification on problems, but it is up to the board's discretion whether to explain.
- The time limit is **30 minutes**.
- There will be 8 questions.
- The test is free-response. Make sure to simplify answers completely. (Hint: you should be getting integers).
- Problems are not necessarily arranged in difficulty order.
- Each correct answer is worth 25 points. The team round is out of 200 points total. There is no penalty for guessing.

Team name: _____

Team members:

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1. Consider triangle ABC with side lengths $AB = 3$, $AC = 5$, and $BC = 7$. Define a *bunny* of a triangle as a circle drawn such that the circle is tangent to the three lines that define the sides of the triangle. If the sum of the areas of the 4 distinct *bunnies* of the triangle ABC can be expressed as $k \cdot \pi$, find $12 \cdot k$.

2. Consider triangle ABC with side lengths $AB = 3$, $AC = 5$, and $BC = 7$. Draw the angle bisector l of $\angle BAC$. Now, tessellate the plane with triangle ABC. Define the vertices of this plane to be the vertices of the tessellated triangles. What is the smallest distance along l between 2 vertices of the plane?

3. Call a number *fat* if the sum of its digits is larger than the sum of its proper divisors. Find the largest *fat* number less than 1000. (A number's proper divisors is its divisors excluding itself.)

4. Find the value of the sum $\sum_{i=1}^{2026} \lfloor \sqrt{i} \rfloor$.

5. Find all ordered pairs (x, y) such that $x + \frac{x+8y}{x^2+y^2} = 2$ and $y + \frac{8x-y}{x^2+y^2} = 0$.

6. There exists a polynomial $P(x)$ of degree 2025 such that $P(k) = D_k$ for integer k from 0 to 2025 inclusive. Find $P(2026)$ in terms of D_{2026}. (Note: D_k represents the number of derangements of k elements. A derangement of n elements is a permutation of the n elements such that the 1st element isn't in the first position, the 2nd element isn't in the second position, and so on.)

7. Define a tetrahedron with edges of length 6, 8, and 12 all perpendicular to each other. Let the radius of the sphere inscribed within this tetrahedron be in the form $\frac{a-b\sqrt{c}}{d}$, where $\gcd(a, d) = 1$, $\gcd(b, d) = 1$, and c is square-free. Find $a + b + c + d$.

8. Find the sum of all values of n such that $0 < n < 100$ and 21 divides $2^{2^n} + 2^n + 1$.